

SAGAR SURESH VATHAR

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PROFILE SUMMARY

An VLSI enthusiast with a strong foundation in Digital circuits, CMOS fundamentals and RTL design concepts. Passionate about semiconductor design and eager to contribute in a challenging environment that allows me to apply my technical knowledge while continuously learning and growing in the field of chip design and verification.

EDUCATION

KLE Technological University, Hubballi B.E. Electrical and Electronics Engineering	9.13 CGPA (Till 6th Sem)	2023-2027
Sainik School Bijapur, Vijayapura Class XII	77.2%	2022
Sainik School Bijapur, Vijayapura Class X	89.8%	2020

CORE COMPETENCIES

- Good Understanding of CMOS Fundamentals.
- Efficient in drawing Schematics for given Boolean equations.
- Efficient in designing digital circuits and implementing the RTL modules.
- Hands-on experience in 45 nm GPDK technology in Cadence Virtuoso.
- Designed transistor-level schematics and custom layouts of CMOS logic gates (INV, NAND, NOR, XOR, MUX), followed by DRC and LVS verification.
- Efficient in Solving DRC Errors (spacing/enclosure/area) & LVS Errors (Shorts & open/instance/pin mismatch/parameter mismatch/ missing nets).

PROJECTS

1. 5-Stage Pipelined MIPS32 Processor | Verilog HDL | FPGA

- Designed and implemented a 5-stage pipelined MIPS32 processor using Verilog HDL.
- Developed the Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), Memory Access (MEM), and Write Back (WB) pipeline stages.
- Implemented the MIPS32 instruction set, Datapath, and control logic.
- Verified functional correctness through behavioural simulation using a MIPS32 assembly program for decimal-to-binary conversion based on shift and divide operations.
- Developed a Python-based assembler to translate the MIPS assembly language into 32-bit machine code.

- Successfully synthesized, implemented, and validated the processor on an FPGA (Spartan 6), with the execution output displayed on an LCD interface.

2. Configurable 1S Battery Protection Unit (BPU) IC | Verilog HDL | ASIC Design (Ongoing)

- Contributing to the digital part of a configurable 1S Battery Protection Unit (BPU) IC for wearable applications.
- Designed and developed the RTL of the Data Acquisition Controller, including the controller unit comprising a Finite State Machine (FSM) and 2-bit counters.
- Integrated the FSM and counter modules to implement the data acquisition control logic.
- Integrated the Controller and Datapath modules to implement the Data Acquisition logic.
- Verified RTL functionality using behavioral simulation and gate-level simulations (post-translate, post-map, and post-route) and validated the design on FPGA.

3. 8:3 Priority Encoder | Cadence Virtuoso | CMOS VLSI Design

- Designed the transistor-level schematic of an 8:3 Priority Encoder using Static CMOS and Pseudo-NMOS logic styles.
- Implemented and compared both logic styles in Cadence Virtuoso.
- Verified the functionality through schematic-level simulation.
- Analysed the trade-offs between Static CMOS and Pseudo-NMOS in terms of power, delay, and area.

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python, Verilog HDL.
- **Tools and Platforms:** Git, Visual Studio Code, ISE Design suite, Vivado, Cadence Virtuoso, Basic Excel.

SOFT SKILLS

- Leadership Skills, Solution-Oriented Approach, Teamwork